Optics Communications 574 (2025) 131129

Contents lists available at ScienceDirect

Optics Communications

journal homepage: www.elsevier.com/locate/optcom

Optimized domain-engineered lithium niobate for electro-optic spectral tuning in a multi-wavelength optical parametric oscillator

Lin-Ming Deng [a], Shue-Shan Lin [a], Tien-Dat Pham [a], Yen-Hung Chen [a,b,c,*]

[a] Department of Optics and Photonics, National Central University, Jhongli, 320, Taiwan
[b] Quantum Technology Center, College of Science, National Central University, Jhongli, 320, Taiwan
[c] Center for Astronautical Physics and Engineering, National Central University, Jhongli, 320, Taiwan

ARTICLE INFO

Keywords:
Nonperiodically poled lithium niobate
Multiline optical parametric oscillator
Electro-optic spectral tuning

ABSTRACT

We report on an electro-optically (EO) spectrum tunable, multiline optical parametric oscillator (OPO) based on a nonperiodically poled lithium niobate (NPPLN). Besides being an efficient optical parametric down converter (OPDC) for multi-wavelength signal generation, the NPPLN can function as an EO modulator for fast spectral tuning based on a unique asymmetric-duty-cycle (ADC) equivalent domain structure. We have developed a genetic algorithm to construct and optimize such an ADC equivalent domain configuration in NPPLN to maximize its EO spectral tuning rate with the least compromise of its nonlinear gain for conducting the multiple down-conversions. When the novel EO NPPLN device with an equivalent ADC of 29%/71% works in an OPO intracavity pumped by a diode-pumped, 1064-nm Nd:YVO_4 laser, we measured EO tuning rates of ~0.5 and ~0.53 nm/(kV/mm) for the down-converted dual signals at 1540 and 1550 nm, respectively. The tuning rates have been three orders of magnitude higher than a conventional one without an engineered domain asymmetry and 12% greater than another domain asymmetric LN device designed by a simulated annealing optimization method. The EO tuning technology developed in this study can be potentially implemented in micro/nano-photonic LN waveguide circuits.

1. Introduction

Fast tunable pulsed coherent light sources are of great value for a wide range of applications including remote sensing, spectroscopy, biological imaging and detection, astronomy, and communications. In contrast to lasers [1], tunable sources built in optical parametric oscillators (OPOs) can be of great versatility as they can be realized in a broadly accessible down-conversion spectral region [2]. On the other hand, tunable sources with multiple emission lines can be of more interest to those aforementioned applications to further increase the system processing speed, precision, and multiplexity [3–6]. Such multiline sources have also become popular in the coherent THz wave generations through the nonlinear-optic difference frequency generation process [7]. Multi-wavelength OPOs have been demonstrated in quasi-phase-matching (QPM) optical parametric down converters (OPDCs) thanks to the high nonlinearity and highly domain engineerable characteristics of the QPM materials [8–11]. In contrast to the approaches of using cascaded or multiple uniform QPM-grating (i.e., periodically poled) structures for the multi-wavelength generation [10, 11], we have shown that a non-uniform domain structure engineered in QPM materials can be more advantageous in terms of the overall conversion efficiency and the power spectral density due to the increased effective conversion length (which also leads to a narrowed conversion bandwidth) [12]. Several methodologies have been proposed to construct a specific domain structure in a QPM material to create the desired multiple conversion channels including the nonperiodic optical superlattice (NOS) [13] and aperiodic optical superlattice (AOS) techniques [14]. The resultant domain grating structure is usually non-uniform (aperiodic) in order to create multiple reciprocal vectors (or spatial frequencies, associated with the Fourier components in the Fourier expansion of the domain structure) to phase-match the desired multiple nonlinear-optic conversion processes.

Spectral tuning of OPOs is usually achieved by varying the phase-matching condition of mixing waves created in the used $\chi^{(2)}$ nonlinear medium to enable the change of the oscillating (phase-matching) wavelengths [15]. To build a spectral tuning mechanism directly in a QPM material-based OPO system, a fan-out and a multiple domain-grating QPM crystals have been used [16,17]. However, such

* Corresponding author. Department of Optics and Photonics, National Central University, Jhongli, 320, Taiwan.
E-mail address: yhchen@dop.ncu.edu.tw (Y.-H. Chen).

https://doi.org/10.1016/j.optcom.2024.131129
Received 2 July 2024; Received in revised form 14 September 2024; Accepted 18 September 2024
Available online 19 September 2024

tuning methods rely on the mechanical translation of the crystals which is slow. Among the optical tuning/modulation/switching approaches, electro-optics (EO) is in general the most favorable one because of its predominantly fast, precise, repeatable, and stable operating properties [18]. The EO (Pockels) effects of QPM materials have been studied to implement several unique fast spectral tailorable, tunable, and switchable OPOs [8,19–21]. In our previous study [20], we achieved EO spectral tuning in an OPO built based on a fan-out-grating periodically poled lithium niobate (PPLN) with a ramped duty-cycle domain. Though having a relatively high tunability, the tuning range of the system in Ref. [20] is highly limited by the geometry of the oscillator, and the ramped domain structure also limits the flexibility of further designing/engineering an efficient multi-wavelength converter. We also have first introduced the concept of using an "asymmetric duty cycle" (ADC) PPLN [22] as a mechanism for EO spectral tuning into an aperiodically poled lithium niobate (APPLN) to achieve a fast tunable multiline source [23]. The EO effect changes the refractive indices and therefore, induces an uncompensated wave-vector mismatch for the mixing waves (originally phase-matched) in such an ADC domain structure [24], leading to the shift of the (down-) converted wavelengths subject to the new (electro-optically modified) phase-matching condition. However, the spectral tuning resolution and fidelity of the source presented in that work [23] are not satisfactory due to the exhibited relatively broad signal bandwidth and the less ability of the employed domain construction method to optimize a multiline nonlinear gain spectrum. Besides a broad gain bandwidth, the single-pass optical parametric generation (OPG) process employed in the system also features relatively high threshold power (where a bulky pulsed pump laser has been used).

In this work, we further propose and demonstrate an EO spectrum tunable, multiline OPO based on a nonperiodically poled lithium niobate (NPPLN) device designed to function both as an OPDC and an EO modulator in an ADC equivalent domain structure. The OPO exhibits several important features, especially in contrast to the previous OPG scheme. First, the domain structure of the NPPLN is calculated by a genetic algorithm whose powerful optimization ability has led the device to implement a multiline OPO with a higher spectral tuning rate and fidelity. Second, the OPO is realized in a diode-pumped solid-state laser, largely reducing the spectral linewidth and the system size.

2. NPPLN design and performance simulation

In this study, we first design an EO tunable, multi-wavelength OPDC based on the domain engineering of a LN chip using the NOS technique for operating in a compact OPO which is intracavity pumped by a solid-state (Nd:YVO_4) laser. The NOS approach is employed to find an ADC equivalent, nonperiodically poled domain structure in LN with the aid of a genetic optimization algorithm for maximizing the EO spectral tuning rate in the OPO under a specific nonlinear gain that still leads to an efficient multiple down-conversion process (the nonlinear gain of an ADC-domain QPM device will be inevitably reduced according to the theory). The EO spectral tuning rate of the NPPLN studied in this work can be derived from the general QPM theory and EO effect, given by

$$\frac{\Delta\lambda_{s,j}}{\Delta E_z} = \frac{\lambda_{s,j}(E_z) - \lambda_{s,j}(0)}{E_z} \tag{1}$$

with $\lambda_{s,j}(E_z) = \left\{\lambda_p\left(n_{s,j} - n_{i,j} + \Delta D_L\left(r_{33,s}n_s^3 - r_{33,i}n_{i,j}^3\right)E_z/2\right)\right\} / \left\{(n_p - n_{i,j}) + \Delta D_L\left(r_{33,p}n_p^3 - r_{33,i}n_{i,j}^3\right)E_z/2 - \lambda_p K_j/2\pi\right\}$, where the subscript j denotes quantities related to the j-th QPM down-conversion process performed in the NPPLN, while p, s, and i are associated to the three mixing waves, i.e., the pump, signal, and idler, respectively, λ is the wavelength, n is the refractive index of the phase-matched waves, r_{33} is the relevant EO coefficient (accessed by an extraordinary wave in LN under an electric field E_z applied along the crystallographic z axis) [25], K is the reciprocal vector created by the domain structure of the NPPLN for satisfying the QPM condition of the relevant nonlinear conversion process, and $\Delta D_L \equiv \left(D_L^+ - D_L^-\right) = (L^+ - L^-)/L$ is the difference of the normalized lengths between the positive and negative domains (of total lengths L^+ and L^-, respectively) that constitute the NPPLN of length L, which is a factor indicating the domain asymmetry of such a device. From Eq. (1), the calculated EO spectral tuning rate is ~0.51 nm/(kV/mm) for a 1064-nm pumped 1550-nm signal and 3393-nm idler OPDC process in a domain structure with a $\Delta D_L = 0.4$.

It can easily be shown from Eq. (1) that the EO spectral tuning rate is mainly proportional to ΔD_L, i.e., the asymmetry between the two-polarity domains constituting the NPPLN, as well as the EO coefficient (r_{33}), as expected. However, as in a conventional QPM device with an ADC domain, the higher the ΔD_L of our multi-wavelength NPPLN device is, the lower the wavelength conversion efficiency will be according to the nonlinear gain coefficient derived from the domain structure:

$$G_j = \frac{1}{L}\left|\int_0^L s(x)e^{-i\int_0^x \Delta k_j' dx}dx\right| = \frac{1}{\Delta k_j' L}\left|1 - (-1)^N e^{-i\Delta k_j' L} + 2\sum_{q=1}^{N}(-1)^q e^{-i\Delta k_j' x_q}\right|, \tag{2}$$

assuming the NPPLN is composed of a number of N domain segments where the signs $s(x)$ of their domain polarity are alternatively varied between +1 and −1, denoting the +z and −z crystal orientations, respectively, along the x axis with domain boundary positions at x_1, x_2, …, and x_{N-1} (while x_0 and x_N denote the positions of the front and back ends of the NPPLN, respectively). In Eq. (2), $\Delta k_j' = k_p - k_{s,j} - k_{i,j}$ is the wave-vector (k) mismatch of the 3 mixing waves in the NPPLN for the j-th OPDC process. Being a QPM scheme, ideally the thickness of each domain segment, $t_q = x_q - x_{q-1}$ for q = 1, 2, …, N, is equal to the coherence length of the wave mixing process, $l_c(\equiv \pi/\Delta k')$, as implied by Eq. (2) that G (=G_{PPLN}) reaches its maximum when $t_q = l_c$. In this ideal case, it reduces to a PPLN with a symmetric domain duty cycle or with $\Delta D_L = 0$. However, in a NPPLN device for multi-wavelength conversions, the domain thicknesses (t_q for q = 1, 2, …, N) will not be uniform to form multiple reciprocal vectors K_j ($\equiv \pi/l_{c,j}$) for satisfying all the QPM conditions of the wave mixing processes subject to their respective coherence lengths $l_{c,j}$. The nonlinear gain of each wave mixing process in a NPPLN will thus be lower than the corresponding case in a PPLN (i.e., $G_j < G_{PPLN}$) as not all the t_q can be equal to a specific $l_{c,j}$. An optimized G_j can be pursued when each t_q can best approach $l_{c,j}$ (so that ΔD_L can be minimized) to maximize the QPM coherence-sum term appearing in Eq. (2). The genetic algorithm developed here can be applied to find an optimized domain structure (i.e., optimized positions x_q) for supporting equalized and maximized nonlinear gains G_j for all the multiple OPDC processes. However, to create and enhance the tunability, a higher ΔD_L is desired as discussed above, which will lead to a lower G_j as the mean domain thicknesses of the positive and negative domain segments, denoted as $\bar{t}^+$ and $\bar{t}^-$, respectively, are neither close to each other (i.e., to form ADC equivalent domains) nor close to $l_{c,j}$ (however, $2l_{c,j} \sim (\bar{t}^+ + \bar{t}^-)$ should still be obtained to form the required periodicities to meet the QPM conditions). In this study, we have developed a genetic algorithm to derive a domain structure in LN with an enhanced ΔD_L but with the least compromise of the nonlinear gain G for building an efficient EO tunable, multiline OPO in the 1.5-μm telecom band. In the algorithm, we first set the number of domain segments, corresponding to the number of genes contained in each chromosome that participates as an individual of the initial population created in the genetic evolution, to be 2670. This number is reasonably determined according to the coherence lengths $l_{c,j}$ of ~15 μm for a 1064-nm pumped 1.5-μm signal-band OPDC process and the device length L of 4 cm. A chip length of 4 cm is employed to allow the creation of a sufficiently large number of variables for increasing the diversity of the gene encoding in chromosomes to facilitate the algorithm to reach a (globally) optimal solution and in addition, to provide a relatively long nonlinear gain length while

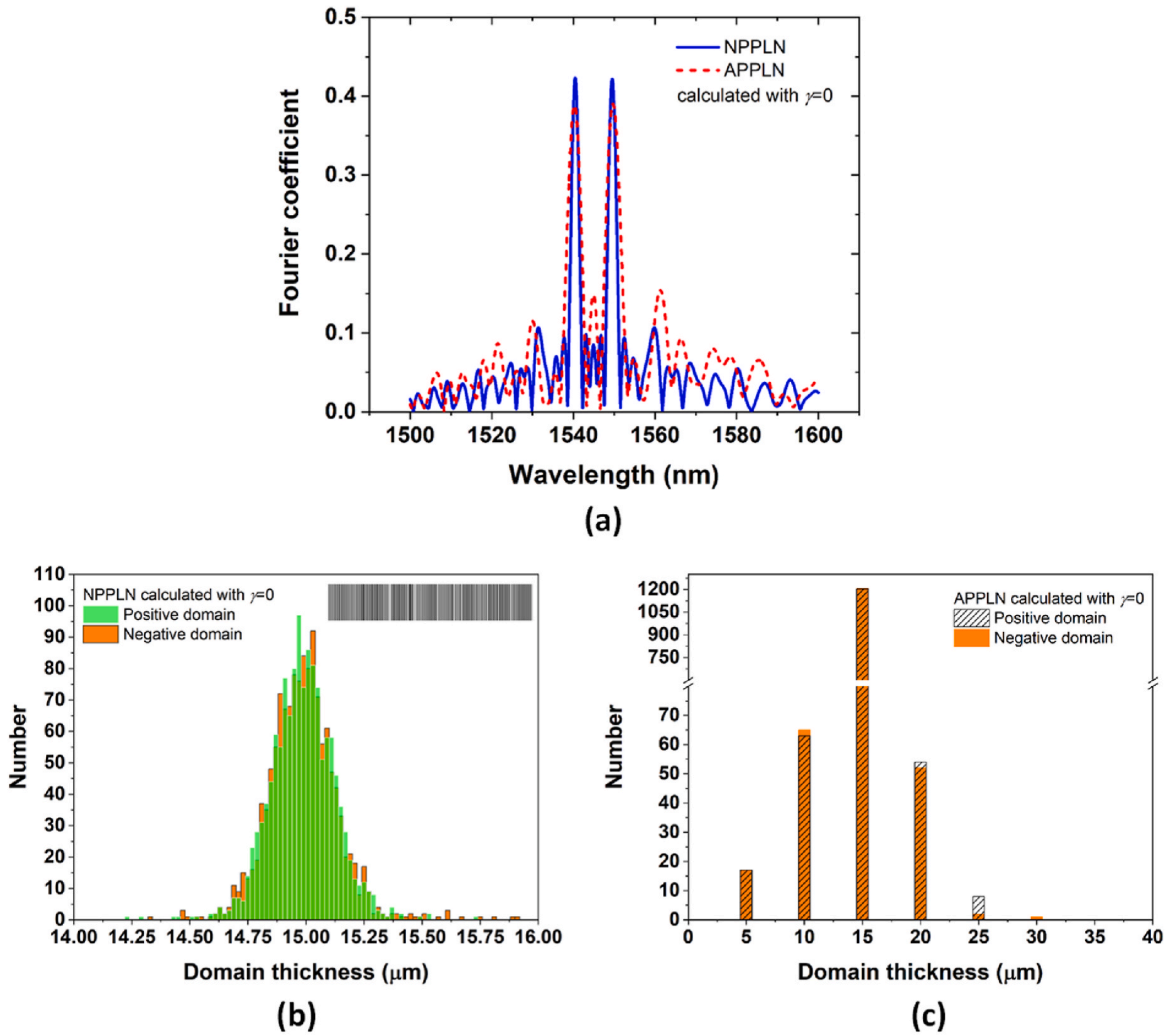

Fig. 1. (a) Fourier spectra of the NPPLN and APPLN dual-signal OPDCs, calculated by the genetic algorithm and SA method, respectively, with $\gamma = 0$. (b) and (c) Distributions of domain thickness analyzed from the domain structures of the calculated NPPLN and APPLN, respectively. The inset of (b) displays the barcode-like domain structure of the NPPLN.

compatible to build a compact all-solid-state tunable coherent light source. The values of the variables, corresponding to the thicknesses of the constituent domains of a NPPLN are first randomly set by the computer within an appropriate range with a resolution of 0.01 μm, in which the initial genetic code for each individual is generated. The gene string coded for each individual (chromosome) thus corresponds to one particular NPPLN domain structure. We further divide the created population into eight subpopulations with each containing 2000 individuals and then perform the genetic evolution procedure with those initial settings. The procedure includes operators such as selection, crossover, mutation, and migration [26], which is an optimization process inspired by the natural selection mechanism to evolve to a new population (offspring) with better fitness. Proper selection, crossover, mutation, and migration rates found for these operations are 20%, 80%, 0.07%, and 0.1%, respectively, in this study. A fitness/objective function (*FF*) has been employed to help converge the evolution process (and therefore the algorithm) to yield an optimum gene string, corresponding to here the NPPLN domain structure, that can lead to achieving a desired tunability while acquiring a maximally accessible nonlinear gain when working in an OPO intracavity pumped by a Nd:YVO$_4$ laser, given by

$$FF = \left\{ \sum_{j=1}^{M} \left[w(\lambda_j) \left| G_t(\lambda_j) - G_c(\lambda_j) \right| \right] \right\} + \beta\{max[G_c(\lambda_1), \ldots, G_c(\lambda_M)] - min[G_c(\lambda_1), \ldots, G_c(\lambda_M)]\} + \gamma \left| \Delta D_{L,t} - \Delta D_{L,c} \right|, \tag{3}$$

where M is the number of multiple OPDC processes performed in the NPPLN, $G_t(\lambda_j)$ and $G_c(\lambda_j)$ are the target and calculated parametric gain coefficients (given in Eq. (2)) of the j-th process for generating a signal at wavelength λ_j, $w(\lambda_j)$ is the weighting factor on gain for the j-th process, β is a factor properly set to achieve equalized peak gains/conversion

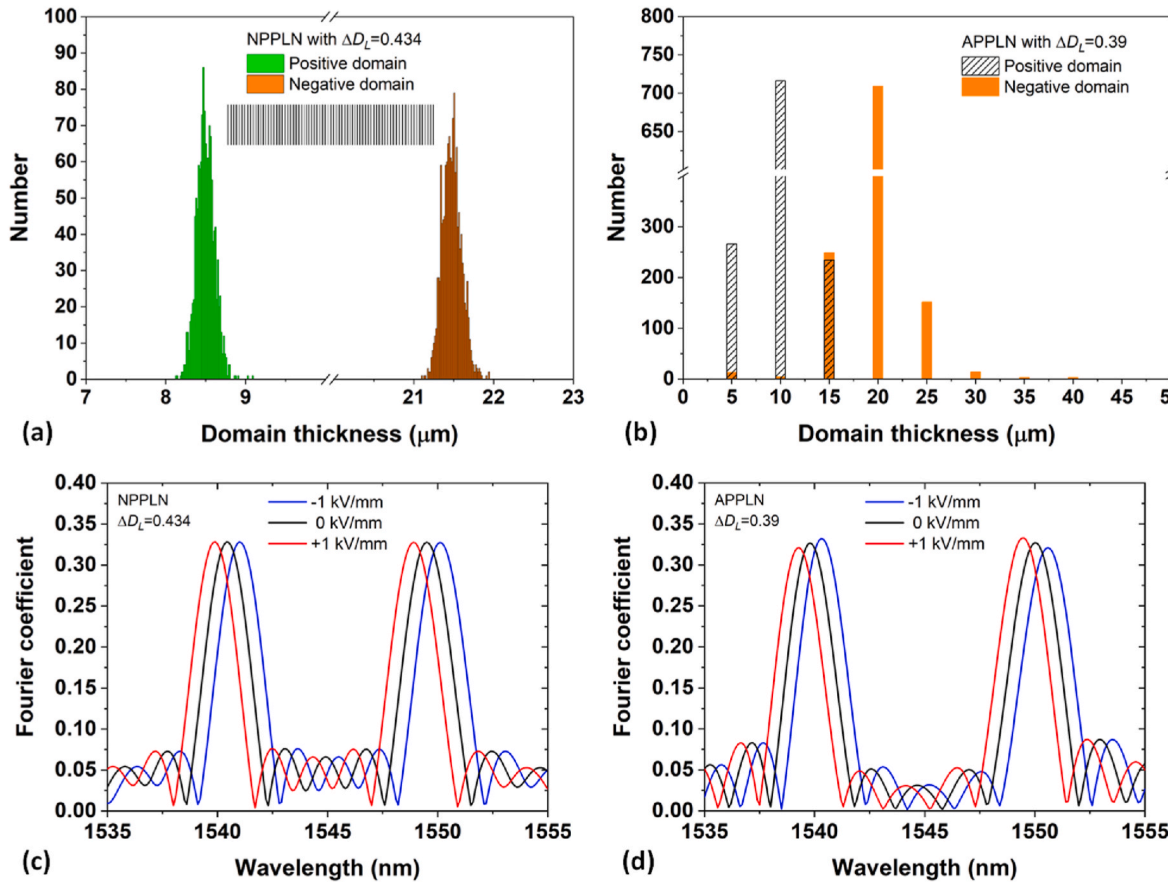

Fig. 2. (a) and (b) Distributions of domain thickness analyzed from the domain structures of the calculated NPPLN and APPLN, respectively, having a maximized tuning rate (ΔD_L) under $G_t = 0.33$. The inset of (a) displays the barcode-like domain structure of the NPPLN with $\Delta D_L = 0.434$. (c) and (d) Fourier (gain) spectra of the NPPLN^{EO} and APPLN^{EO} dual-signal OPDCs when operating at $E_z = -1$, 0, and $+1$ kV/mm, respectively.

efficiencies for the $\boldsymbol{M}$ down-conversion processes with the aid of the operators max[…] and min[…] which select the maximum and minimum values from the quantities enclosed in the square brackets, $\Delta D_{L,t}$ and $\Delta D_{L,c}$ are the target and calculated ΔD_L (the asymmetry between the two-polarity domains), and γ is the weighting factor on this domain asymmetry (which is associated to the tuning rate). The optimization procedure will be repeated until either the *FF* returns a value lower than or equal to the set one (which is 0 in this work) or the iteration number reaches a set upper limit (which is 10000, corresponding to the number of generations, in our case). Finally, the chromosome with the minimum fitness value produced through the above genetic evolution procedure gives the information of an optimal NPPLN domain structure.

We first calculate a NPPLN that can perform as a dual-signal OPDC producing wavelengths at $\lambda_1 = 1540$ and $\lambda_2 = 1550$ nm (and corresponding idlers at 3442 and 3393 nm) at 40°C when pumped by a 1064-nm wave using the developed genetic algorithm without considering any tunability (i.e. when $\gamma = 0$ is used in Eq. (3)). Fig. 1(a) shows the gain (Fourier) spectrum (blue curve) of the NPPLN OPDC, calculated from Eq. (2) with the domain structure constructed and optimized by the genetic algorithm. The calculated Fourier spectrum (red dashed curve) of an APPLN OPDC is also plotted for comparison. Both devices have a length of 4 cm. The APPLN domain structure is obtained by an algorithm we have developed previously [23] based on the simulated annealing (SA) optimization method [27]. It clearly shows the NPPLN can provide equalized dual-signal gains with a higher peak value and a lower background level (side lobes) in contrast to those with the APPLN. The superior optimization power of a genetic algorithm over a SA method can be expected as the thicknesses of the domains used to constitute a NPPLN can be arbitrary values (can be set randomly in the calculation as aforementioned) within a set range in contrast to that in an APPLN where the constituent domains are restricted to have thicknesses a multiple of that of a set unit domain building block (UDB) [14]. Fig. 1(b) and (c) show the distributions of domain thickness analyzed from the domain structures of the calculated NPPLN and APPLN, respectively, where the green (or shadowed) and orange bars denote the positive and negative domain polarities, respectively. The inset displays the barcode-like domain structure of the NPPLN with black and white representing the two opposite polarities. It clearly shows the domain thicknesses (t_q) are distributed continuously in a set resolution of 0.01 μm for the NPPLN, while in the APPLN, they are distributed only at specific values that are multiples of 5 μm (which is the thickness of the UDB, set according to the minimum domain thickness we can pole in currently established technology). In the case of NPPLN, we can find that the distribution peaks around 15 μm and has an average of ~14.98 μm which is exactly the average of the coherence lengths associated to the two 1064-nm pumped OPDC processes in LN at 40°C. It can be seen that the amounts of the positive and negative domains are about the same; the length difference between the two domains (i.e., L^+ and L^-) is calculated to be 4.97 μm, corresponding to a domain asymmetry $\Delta D_L =$ 0.000124, which leads to an EO spectral tuning rate of ~0.000157 nm/(kV/mm) at 1550 nm according to Eq. (1). The tuning rate is very limited as expected, as the domain calculation only considers the optimization of the nonlinear conversion (OPDC) efficiency. Similar tunability can be found in the APPLN counterpart.

To enhance the tunability, as discussed above, the domain asymmetry should be increased with, however, a compromise of the OPDC nonlinear gain. We next pursue here with the genetic algorithm to find a maximized tuning rate (proportional to ΔD_L) under a certain parametric

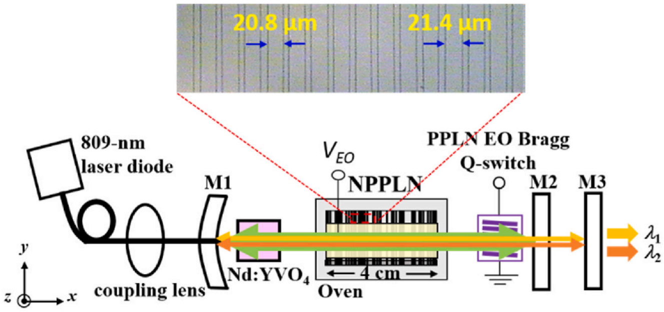

Fig. 3. Schematic arrangement of an EO tunable, dual-line OPO system intracavity pumped by a diode-pumped, 1064-nm $Nd:YVO_4$ laser based on the developed $NPPLN^{EO}$ OPDC. The inset shows an image of a portion of the HF-etched + z surface of the $NPPLN^{EO}$ chip. The measured thicknesses of the negative domain segments as marked in the image indicate a result of an under-poled domain structure.

gain with the aid of the *FF* (Eq. (3)). Fig. 2 shows the corresponding results when calculated with $G_t = 0.33$. This gain level is set by referring to our previous experimental experience in building an efficient multi-wavelength OPO system [8]. Other parameters used in the *FF* to expedite the algorithm to reach the optimized result are found to be $w(\lambda_1) = w(\lambda_2) = 1$, $\beta = 0.3$, $\gamma = 0.1$, and $\Delta D_{L,t} = 0.5$. It is interesting to observe from Fig. 2(a) that t_q of the NPPLN are now distributed into two groups with mean thicknesses being at $\bar{t}^+ = 8.48$ μm and $\bar{t}^- = 21.48$ μm, respectively. As expected, the average of $\bar{t}^+$ and $\bar{t}^-$, which is 14.98 μm, is just the average of the coherence lengths of the two OPDC processes. Such a NPPLN (referred to as $NPPLN^{EO}$ hereinafter) domain structure has a ΔD_L~0.434, enabling an EO spectral tuning rate of 0.55 nm/(kV/mm) at 1550 nm or >3500-fold higher than the NPPLN calculated with $\gamma = 0$ (referred to as $NPPLN^0$ hereinafter). Fig. 2(c) shows the Fourier spectra of the $NPPLN^{EO}$ OPDC when operating at $E_z = -1$, 0, and +1 kV/mm, respectively. The corresponding spectra of an APPLN (referred to as $APPLN^{EO}$ hereinafter) calculated using the SA algorithm under the same target parametric gain and tunability are also presented (see Fig. 2(d)) as a comparison. It can be found that the peak gain coefficients (G_c) of the $NPPLN^{EO}$ dual signals do approach the target $G_t = 0.33$ and remain equalized with respect to the electric-field tuning (from −1 to +1 kV/mm). In contrast, equalized dual-signal gain spectra can't be maintained with the $APPLN^{EO}$ OPDC during the EO tuning. Such a loss of spectral fidelity could be attributable to a higher limitation on configuring an optimized device domain with the AOS technique as, first, it requires a certain difference in numbers between the positive and negative UDBs in the domain construction to enable a certain tunability and, second, it needs the thicknesses of the constituent domains to be a multiple of that of the UDB as discussed above, which limits the ability to optimize the operational bandwidth of the device. The spectral fidelity can be even degraded when the multi-wavelength devices with such an uneven-peak Fourier spectrum experience a parametric amplification in an OPG/OPO due to the nonlinear gain [23]. The EO spectral tuning results also indicate the $APPLN^{EO}$ device reaches a tuning rate of 0.49 nm/(kV/mm) at 1550 nm, implying a domain structure having a ΔD_L~0.39. The domain distribution of such an $APPLN^{EO}$ has been shown in Fig. 2(b).

Therefore, besides the superiority in the EO spectral tuning fidelity, the created $NPPLN^{EO}$ device further features a higher EO tuning rate (enhanced by 12%) in contrast to the $APPLN^{EO}$ device under the same OPDC gain according to the results shown in Fig. 2, showing the great effectiveness of our developed genetic algorithm.

3. NPPLN device fabrication, IOPO construction, and output performance characterization

To demonstrate the proposed multi-wavelength EO tuning

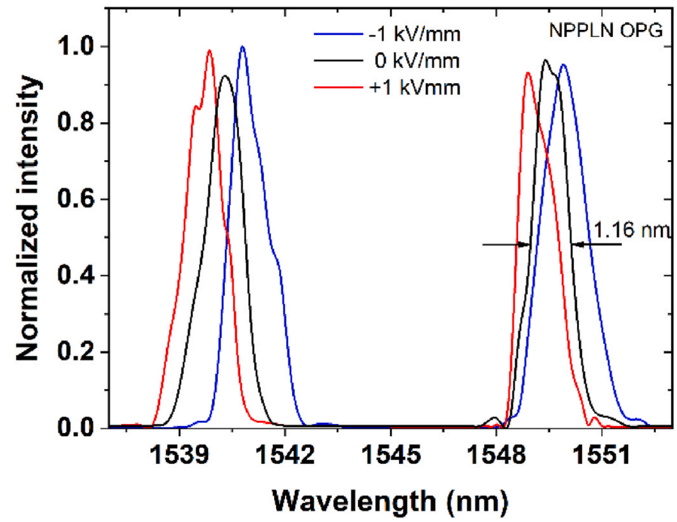

Fig. 4. Spectral tuning of the OPG when the $NPPLN^{EO}$ chip is applied with voltages at −500, 0, and +500 V (corresponding to $E_z = -1$, 0, and +1 kv/mm), respectively, at 40°C under 6.4-W diode pump power.

technology, we fabricated the $NPPLN^{EO}$ device in a 40-mm-long, 6-mm-wide, and 0.5-mm-thick z-cut LN chip by first translating the calculated barcode-like domain pattern onto the $+z$ surface of the chip as a photomask for the subsequent electric-field domain poling [28], followed by doing the polishing and anti-reflection coating (with transmittance of >99.1% at 1064 nm and >99.3% at 1500–1580 nm; the transmittance at the nonresonant idler (~3.4 μm) is found to be ~52.2%) on the two end (x) faces, and finally electrode (Au/Ti) coating on both of the crystal z surfaces for the E_z application. The fabricated $NPPLN^{EO}$ was then mounted on a temperature-controlled crystal oven and installed in an OPO intracavity pumped by a diode-pumped, EO Q-switched $Nd:YVO_4$ laser. Fig. 3 shows the schematic arrangement of such an EO tunable, dual-line OPO system realized by using the unique $NPPLN^{EO}$ OPDC. The inset shows an image of a portion of the HF-etched + z surface of the $NPPLN^{EO}$ chip, revealing an under-poled domain structure (the effect will be discussed below). In the system, a 9-mm long, a-cut 0.3-at. % $Nd:YVO_4$ crystal with a 3 mm × 3 mm clear aperture is used as the laser gain medium and is installed in a water-cooled heat sink. The $Nd:YVO_4$ laser, pumped by an 809-nm fiber coupled laser diode, oscillates at 1064 nm in a resonator formed by a 15-cm radius-of-curvature meniscus BK7 mirror (M1) and a plane-plane BK7 mirror (M2). M1 has ~93% transmittance at 809 nm and ~99.8% and >99% reflectance at 1064 and 1510–1580 nm, respectively, while M2 has ~99.6% reflectance at 1064 nm and >95% transmittance at 1510–1580 nm. A homemade PPLN EO Bragg cell [29] (having the same coating specifications as the $NPPLN^{EO}$ chip), driven by a 170-V, 300-ns voltage pulse train at 1 kHz, is used as a fast (ns) Q switch in the system to produce high peak-power fundamental laser pulses to efficiently drive an intracavity OPO (IOPO). The IOPO, formed by M1 and an output coupler (M3), is configured to resonate at the dual-signal lines produced from the $NPPLN^{EO}$ OPDC chip intracavity pumped by the 1064-nm laser. M3 is a plane-plane BK7 mirror having ~94% reflectance at 1510–1580 nm. The low (<20%) transmittance of the mirror material (BK7) at the (mid-IR) idler band permits the system to oscillate at the signals only to prevent possible instability issues found in a doubly-resonant scheme [30]. We don't observe an effect of the mirror absorption of the idler wave on the IOPO system, which might be due to a relatively low operating power level (on the order of 10 mW on average; see below) of our system in this proof-of-principle demonstration experiment. To free from the possible mirror absorption issues, one can resort to the replacement of the BK7 cavity mirrors with such as ZnSe or MgF_2 mirrors transparent to a broader IR range. The nonlinear frequency

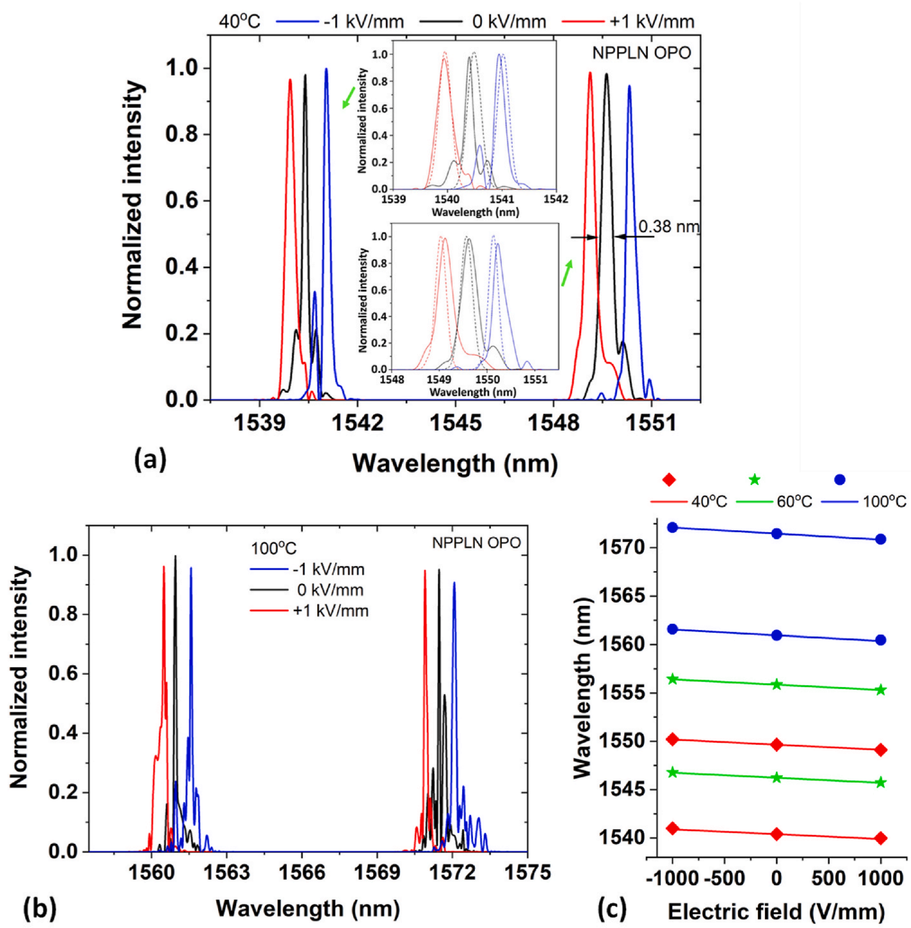

Fig. 5. Spectral tuning of the dual-line IOPO when the NPPLN^{EO} chip is applied with voltages at −500, 0, and +500 V, respectively, at **(a)** 40°C and **(b)** 100 °C under 6.4-W diode pump power. The inset of **(a)** shows the detailed EO tuning spectra and their theoretical fits. **(c)** Measured and fitted EO tuning rates of the IOPO system as a function of the wavelength at 40°C, 60°C, and 100°C.

down-conversion through the NPPLN^{EO} device provides an efficient mechanism of extracting the laser energy accumulated during the Q-switching operation in a high finesse laser cavity (of high-reflectivity mirrors) to accomplish the generation of fast laser pulses.

The performance of the single-pass conversion scheme (i.e., an OPG) of the built system is first tested when M3 is absent. Fig. 4 shows the output OPG spectra measured from the system when the NPPLN^{EO} chip is applied with voltages at −500, 0, and +500 V (corresponding to $E_z = -1$, 0, and +1 kv/mm), respectively, at 40°C under 6.4-W diode pump power (stable operation of the OPG can only be observed at a pump power >~5.8 W, the threshold power). The results show EO tuning rates of ~0.49 and ~0.52 nm/(kV/mm) at the two signal bands (1540 and 1550 nm) have been obtained from the realized NPPLN^{EO} chip, respectively, implying a domain asymmetry of the fabricated chip to be ΔD_L~0.41, which is slightly less than the design value (ΔD_L~0.434) and can be attributed to the observed under-poled domain structure (see the inset of Fig. 3). From this experimentally derived domain asymmetry, we can deduce that our NPPLN^{EO} device is with an average domain under-poled error of ~1.67% [31].

To enhance the spectral tuning resolution and operate in an efficient system, we test the NPPLN^{EO} chip in an IOPO scheme as presented in Fig. 3. It is expected both the signal linewidth and the pump threshold will be significantly reduced in an OPO in contrast to a single-pass OPG scheme. Fig. 5(a) shows the output spectra measured from the constructed IOPO system when the NPPLN^{EO} chip is applied with voltages at −500, 0, and +500 V, respectively, at 40°C under 6.4-W (the maximum available) diode pump power. The inset shows the detailed EO tuning spectra around the dual signal bands together with their theoretical fits done by a calculation model built based on the coupled-wave theory [19]. The existence of slight wavelength shifts between the measured and calculated spectra could be attributed to our assumption made in the calculation that the OPO cavity condition such as the losses are uniform to all the spectral components considered in the calculation, while the measured spectra reflect a result of the gain competition among the modes in the real OPO cavity condition which should be still dispersive from an engineering perspective. It is found that the spectral linewidths of the dual signals (~0.38 nm) have been well narrowed by a factor of ~3 than that from the OPG (Fig. 4), inferring a transit of approximately nine cavity round trips for the signal buildup [32]. The narrowed signal lines have also led to a much improved EO spectral tuning resolution (narrower than the International Telecommunication Union (ITU) 50-GHz grids at the 1550-nm band), relative to the OPG results, and revealed EO tuning rates to be ~0.5 and ~0.53 nm/(kV/mm) at 1540 and 1550 nm, respectively. A ΔD_L~0.42, derived from the OPO tuning rates, again reflects the fabricated chip is characterized by an under-poled domain structure. Furthermore, the presented EO tuning technology can work over a broad spectral range with the aid of the thermo-optic (TO) effect. Fig. 5(b) shows the EO spectral tuning with the IOPO system when operating the NPPLN^{EO} chip at 100°C under 6.4-W diode pump power. The dual signals are red shifted to around 1561 and 1571 nm with TO tuning rates of ~0.35 and ~0.36 nm/°C, respectively. The corresponding EO tuning rates are ~0.57 and ~0.61 nm/(kV/mm) for the dual signals at 100°C, increasing with the red shifting of the spectrum as predicted by Eq. (1). Since a typical

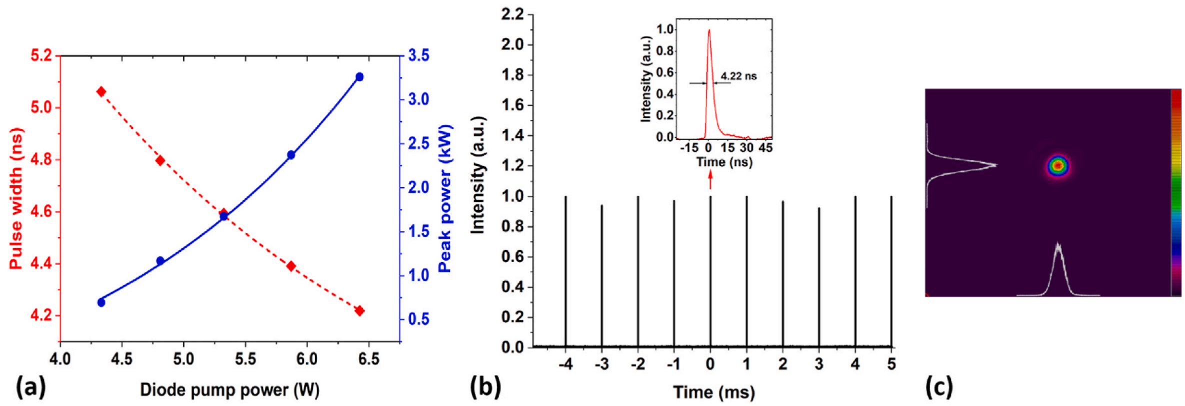

Fig. 6. Output performance of the realized EO tunable, dual-wavelength IOPO system. (a) Measured output peak powers and pulse widths of the system as a function of the diode pump power. (b) Temporal output of the system at 6.4-W pump power. (c) Spatial mode intensity distribution of the output beam of the system at 6.4-W pump power.

temperature controller can easily operate over a temperature difference exceeding 100° with a tuning step of ~0.1 °C, we can use the TO approach to attain a much broader tuning range, though the tuning is coarse and slow in contrast to the EO effect that can provide a much faster and finer tunability. Fig. 5(c) shows the measured and fitted EO tuning rates of the novel IOPO system as a function of the wavelength at several different operating temperatures. All are in good agreement with the calculations. The stability of the output spectra of the IOPO system under repeated EO tunings can be limited by the performance of the employed crystal temperature controller with a ~±0.1°C stabilizability and the voltage supplier with a <0.2 V p-p fluctuation, leading to a spectral shift of <30 pm and a tuning rate change of <0.12 pm/(kV/mm), which has been below the resolution limit (~60 pm) of our spectral analyzer (Agilent 86142B).

The built diode pumped, EO tunable, dual-wavelength IOPO can work at a relatively low pump power of 4.3 W (a level <~4.3 W can then be deemed as the threshold pump power of the system). Fig. 6(a) shows the measured output peak powers and pulse widths of the realized IOPO system as a function of the diode pump power. An output peak power of ~3.3 kW can be obtained from such a system when pumped at (cw) 6.4 W, corresponding to an optical-to-optical (from 808-nm diode to the dual-wavelength signals at 1.5 μm band) conversion efficiency of ~0.22%. Fig. 6(b) shows the typical temporal behavior of the system output, indicating the signal (dual wavelengths) pulses having a pulse width of ~4.22 ns and a peak-to-peak intensity fluctuation of ~7.6%. At this pump level, the output signal beam quality of our IOPO is characterized to be M^2~1.5, determined using the beam profiling method [33] with the aid of an Infrared CCD camera. In this measurement, we used a biconvex lens with a focal length of f = 160 mm to focus the output signal beam to characterize the variation of the transverse beam profiles along the propagation direction. We have acquired beam profile data at 20 different positions along the axis of propagation in the focal region of −1 cm to 1 cm to derive the M^2 of the output signal beam according to the formula $M^2 = \pi d_0 D_s / 4\lambda f$ [33] with d_0~ 602 μm and D_s~ 794 μm being the beam waist spot sizes measured behind and before the lens, respectively, and λ being the signal wavelength (1540/1550 nm when the IOPO operates at 40°C). Fig. 6(c) shows the mode intensity distribution of the beam captured on the focal plane. We believe the beam quality of our source can be further improved by minimizing the width dimension (currently 6-mm wide) of the $NPPLN^{EO}$ chip to reduce the higher order modes and by applying a more careful design for the IOPO cavity configuration. No obvious difference in the output performance of the system is found when performing the EO tuning under the same pump power.

LN is an iconic photonic material widely used to implement efficient nonlinear-optical and electro-optical devices in the form of integrated optics, Currently, the major limitation of the multi-wavelength EO spectral tuning technology developed in this study could be its already high operating voltages (several hundred volts). To further remarkably improve the EO tunability of the demonstrated scheme, the realization of the $NPPLN^{EO}$ OPO in integrated photonic circuits (IPC) like in micrometric LN bulk waveguides [34] and nanometric LN on insulator waveguides [35] is a potential solution to drastically reduce the tuning voltage (by more than two orders of magnitude) as well as the footprint of the proposed device thanks to the greatly size-reduced, well-confined optical modes in the IPC structures. The method developed in this work can potentially also be applied to other QPM materials such as KTP, $LiTaO_3$, GaAs, and GaP.

4. Conclusion

We have realized a unique domain-engineered LN chip designed by a genetic optimization algorithm based on an ADC-domain equivalent mechanism to demonstrate an EO tunable, dual-wavelength OPO constructed in a compact diode-pumped, EO Q-switched $Nd:YVO_4$ laser. The core NPPLN device has a design domain asymmetry of ΔD_L~0.434, enabling an EO spectral tuning rate of 0.55 nm/(kV/mm) at 1550 nm, and a reduced nonlinear coefficient of $G_t = 0.33$. The tuning rate has been drastically enhanced by a factor of 3500 over that with a conventional NPPLN and has been also 12% greater than the APPLN counterpart. When the $NPPLN^{EO}$ chip was applied with an external electric field, we obtained spectral tuning rates of ~0.5 and ~0.53 nm/(kV/mm) for the two oscillating signal waves at 1540 and 1550 nm at 40°C and ~0.57 and ~0.61 nm/(kV/mm) at 1561 and 1571 nm at 100°C, respectively, from the IOPO system. The system thus features a broad spectrum tunable range and a fast and fine (in ITU grid) tunability based on TO and EO controls, respectively, which is attractive to various applications, especially when extended to an IPC platform.

CRediT authorship contribution statement

Lin-Ming Deng: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Shue-Shan Lin:** Writing – review & editing, Validation, Investigation, Data curation. **Tien-Dat Pham:** Writing – review & editing, Software, Formal analysis. **Yen-Hung Chen:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to acknowledge the funding support from the National Science and Technology Council (NSTC), Taiwan under contracts 112-2119-M-008-007, 111-2627-M-008-001, 111-2923-E-008-001, and 111-2923-M-032-002-MY5. The authors thank the Taiwan Semiconductor Research Institute (TSRI), Taiwan, the Nano Facility Center at National Yang Ming Chiao Tung University, Taiwan, and the Instrumentation Center at National Tsing Hua University, Taiwan for the support of the microfabrication facility.

References

[1] S. Wang, H. Liu, W. Mu, Y. Ren, Z. Jia, X. Fu, X. Sun, Y. Jia, Dual-wavelength self-Q-switched mode-locked waveguide lasers based on Nd:LGGG cladding waveguides, Opt. Mater. Express 12 (2022) 854–862.
[2] K.C. Burr, C.L. Tang, M.A. Arbore, M.M. Fejer, Broadly tunable mid-infrared femtosecond optical parametric oscillator using all-solid-state-pumped periodically poled lithium niobate, Opt. Lett. 22 (1997) 1458–1460.
[3] Y. Jiang, S. Karpf, B. Jalali, Time-stretch LiDAR as a spectrally scanned time-of-flight ranging camera, Nat. Photonics 14 (2020) 14–18.
[4] I. Mattis, A. Ansmann, D. Müller, U. Wandinger, D. Althausen, Dual-wavelength Raman lidar observations of the extinction-to-backscatter ratio of Saharan dust, Geophys. Res. Lett. 29 (2002) 20–21, 20-4.
[5] L. Barreau, A.D. Ross, S. Garg, P.M. Kraus, D.M. Neumark, S.R. Leone, Efficient table-top dual-wavelength beamline for ultrafast transient absorption spectroscopy in the soft X-ray region, Sci. Rep. 10 (2020) 5773.
[6] C. Kim, L.V. Wang, Multi-optical-wavelength ultrasound-modulated optical tomography: a phantom study, Opt. Lett. 32 (2007) 2285–2287.
[7] K. Kawase, T. Hatanaka, H. Takahashi, K. Nakamura, T. Taniuchi, H. Ito, Tunable terahertz-wave generation from DAST crystal by dual signal-wave parametric oscillation of periodically poled lithium niobate, Opt. Lett. 25 (2000) 1714–1716.
[8] W.K. Chang, H.P. Chung, Y.Y. Chou, R. Geiss, S.D. Yang, T. Pertsch, Y.H. Chen, Electro-optically spectrum narrowed, multiline intracavity optical parametric oscillators, Opt Express 24 (2016) 28905–28914.
[9] Y.H. Chen, W.K. Chang, H.P. Chung, B.Z. Liu, C.H. Tseng, J.W. Chang, Tunable, pulsed multiline intracavity optical parametric oscillator using two-dimensional MgO:periodically poled lithium niobate-aperiodically poled lithium niobate, Opt. Lett. 38 (2013) 3507–3509.
[10] T. Hatanaka, K. Nakamura, T. Taniuchi, H. Ito, Dual signal-wave optical parametric oscillator using periodically poled $LiNbO_3$ with series gratings, Electron. Lett. 36 (2000) 1409–1411.
[11] V. Ramaiah-Badarla, S.C. Kumar, M. Ebrahim-Zadeh, Fiber-laser-pumped, dual-wavelength, picosecond optical parametric oscillator, Opt. Lett. 39 (2014) 2739–2742.
[12] C.L. Chang, Y.H. Chen, C.H. Lin, J.Y. Chang, Monolithically integrated multi-wavelength filter and second harmonic generator in aperiodically poled lithium niobate, Opt Express 16 (2008) 18535–18544.
[13] J.Y. Lai, Y.J. Liu, H.Y. Wu, Y.H. Chen, S.D. Yang, Engineered multiwavelength conversion using nonperiodic optical superlattice optimized by genetic algorithm, Opt Express 18 (2010) 5328–5337.
[14] B.Y. Gu, B.Z. Dong, Y. Zhang, G.Z. Yang, Enhanced harmonic generation in aperiodic optical superlattices, Appl. Phys. Lett. 75 (1999) 2175–2177.
[15] J.A. Giordmaine, R.C. Miller, Tunable coherent parametric oscillation in $LiNbO_3$ at optical frequencies, Phys. Rev. Lett. 14 (1965) 973–976.
[16] P.E. Powers, T.J. Kulp, S.E. Bisson, Continuous tuning of a continuous-wave periodically poled lithium niobate optical parametric oscillator by use of a fan-out grating design, Opt. Lett. 23 (1998) 159–161.
[17] L.E. Myers, R.C. Eckardt, M.M. Fejer, R.L. Byer, W.R. Bosenberg, Multigrating quasi-phase-matched optical parametric oscillator in periodically poled $LiNbO_3$, Opt. Lett. 21 (1996) 591–593.
[18] O. Svelto, Principles of Lasers, Springer Science+Business Media, New York, 2010 chap. 8.
[19] H.P. Chung, W.K. Chang, C.H. Tseng, R. Geiss, T. Pertsch, Y.H. Chen, Electro-optically spectrum tailorable intracavity optical parametric oscillator, Opt. Lett. 40 (2015) 5132–5135.
[20] W.K. Chang, H.P. Chung, Y.Y. Lin, Y.H. Chen, Electro-optic spectral tuning in a fan-out double-prism domain periodically poled lithium niobate intracavity optical parametric oscillator, Opt. Lett. 41 (2016) 3904–3907.
[21] P. Gross, M.E. Klein, H. Ridderbusch, D.H. Lee, J.P. Meyn, R. Wallenstein, K. J. Boller, Wide wavelength tuning of an optical parametric oscillator through electro-optic shaping of the gain spectrum, Opt. Lett. 27 (2002) 1433–1435.
[22] Y.Q. Lu, J.J. Zheng, Y.L. Lu, N.B. Ming, Z.Y. Xu, Frequency tuning of optical parametric generator in periodically poled optical superlattice $LiNbO_3$ by electro-optic effect, Appl. Phys. Lett. 74 (1999) 123–125.
[23] Y.H. Chen, H.P. Chung, W.K. Chang, H.T. Lyu, J.W. Chang, C.H. Tseng, Electro-optically tunable, multi-wavelength optical parametric generators in aperiodically poled lithium niobates, Opt Express 20 (2012) 28989–29001.
[24] N. O'Brien, M. Missey, P. Powers, V. Dominic, K.L. Schepler, Electro-optic spectral tuning in a continuous-wave, asymmetric-duty-cycle, periodically poled $LiNbO_3$ optical parametric oscillator, Opt. Lett. 24 (1999) 1750–1752.
[25] K.K. Wong (Ed.), Properties of Lithium Niobate, The Institution of Engineering and Technology, London, 2002.
[26] M. Mitchell, An Introduction to Genetic Algorithms, MIT Press, Cambridge, 1996.
[27] S. Kirkpatrick, C.D. Gelatt, M.P. Vecchi, Optimization by simulated annealing, Science 220 (1983) 671–680.
[28] L.E. Myers, Quasi-Phasematched Optical Parametric Oscillators in Bulk Periodically Poled Lithium Niobate, Stanford University, 1995. Ph.D. Dissertation.
[29] J.W. Chang, H.F. Yau, H.P. Chung, W.K. Chang, Y.H. Chen, Characterization and analysis of finite-beam Bragg diffraction in periodically poled lithium niobate electro-optic grating, Appl. Opt. 53 (2014) 5312–5321.
[30] J. Jose, B. Deb, S.D. Gupta, Hopping of chaotic dynamics in a doubly resonant parametric oscillator, Pramana - J. Phys. 49 (1997) 389–398.
[31] Y.W. Lee, F.C. Fan, Y.C. Huang, B.Y. Gu, B.Z. Dong, M.H. Chou, Nonlinear multiwavelength conversion based on an aperiodic optical superlattice in lithium niobate, Opt. Lett. 27 (2002) 2191–2193.
[32] S.J. Brosnan, R.L. Byer, Optical parametric oscillator threshold and linewidth studies, IEEE J. Quant. Electron. 15 (1979) 415–431.
[33] A.E. Siegman, Defining, measuring, and optimizing laser beam quality. Proc. SPIE 1868, Laser Resonators and Coherent Optics: Modeling, Technology, and Applications, 1993.
[34] M. Asobe, Y. Nishida, O. Tadanaga, H. Miyazawa, H. Suzuki, Wavelength conversion using quasi-phase matched $LiNbO_3$ waveguides, IEICE Trans. Electron. 88-C (2005) 335–341.
[35] J. Lu, A.A. Sayem, Z. Gong, J.B. Surya, C.L. Zou, H.X. Tang, Ultralow-threshold thin-film lithium niobate optical parametric oscillator, Optica 8 (2021) 539–544.